Simple interest

1 If $\$16\,716$ of simple interest was earned in 7 years, find how much interest was earned:

a. Each year

b. Each month

2 Assuming that a year has 365 days and 52 weeks, calculate the simple interest earned on the following investments:

a. $\$2540$ at 9% p.a. for 2 years.

b. $\$2010$ at a rate of 6% p.a. for 13 months.

c. $\$1050$ at a semiannual rate of 1.1% for 9 years.

d. $\$7000$ at 1.8% per quarter for 9 years.

e. $\$5320$ at 6% p.a. for 95 weeks.

f. $\$5440$ at 6% p.a. for 566 days.

3 Assuming that a year has 365 days and 52 weeks, calculate the simple interest charged on the following loans:

a. $\$8000$ at 8% p.a. for 6 years.

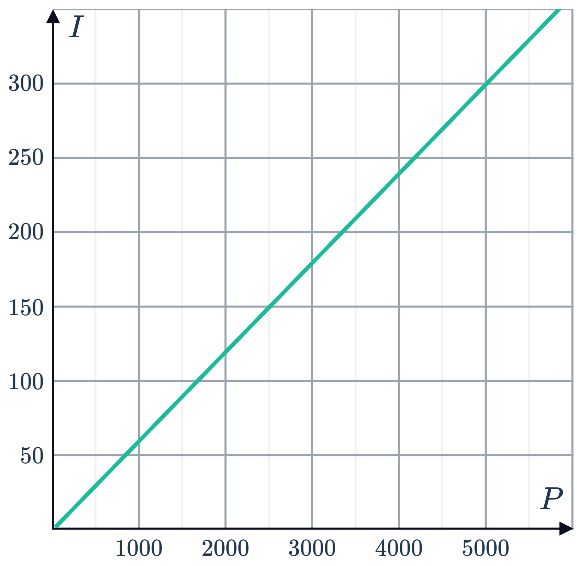
b. $\$3860$ at a rate of 9% p.a. for 13 months.

c. $\$5240$ at 4% p.a. for 73 weeks.

d. $\$5010$ at 5% p.a. for 666 days.

4 Bianca takes out a loan of $\$800$ to pay for an online course. Simple interest is calculated at 9% per year, charged monthly. If she repays the loan in 9 months, how much interest does she pay in total?

5 The graph shows the amount of simple interest charged each year by a particular bank, on a 3-year loan:



a. Find the total amount of simple interest charged on a loan of $\$5000$.

b. Calculate the simple interest rate per annum, r, charged by the bank on 3-year loans.

6 Dave and Emily are looking to invest $\$11\,600$ and $\$18\,300$ respectively for 6 years. The table shows the simple interest rates for different principals:

Principal	Rate p.a.
Between $\$4\,800$ and $\$14\,800$	4%
Between $\$14\,800$ and $\$24\,800$	5%
Greater than $\$24\,800$	6%

a Calculate the interest Dave will earn on his investment.

b Calculate the interest Emily will earn on her investment.

c How much will they earn if they combine their principals and invest together?

d Should they invest separately or together? Explain your answer.

7 Luke's investment of $\$3000$ earned simple interest of 4% p.a. for the first 8 years, and 2% p.a. for the next 5 years.

Calculate the total amount of interest earned.

Final value

8 Calculate the final value of these investments:

a $\$6170$ at 5% p.a. for 6 years.

b $\$9060$ at 4% p.a. for 44 months.

c $\$4230$ at 8% p.a. for 32 weeks.

d $\$4620$ at 6% p.a. for 146 days.

9 $\$402$ is invested at 4% p.a. simple interest for 9 years. After this time the principal plus interest is reinvested at 6% p.a. simple interest for 8 more years.

a Calculate the value of the investment after 9 years.

b Calculate the final value of the investment.

c Calculate the total amount of interest earned.

Principal, rates and time

10 The simple interest earned on each investment is given below. Find the annual interest rate r, as a percentage, correct to one decimal place:

a $\$3600$ over 2 years is $\$504.00$.

b $\$1600$ over 3 quarters is $\$103.20$

c $\$4800$ over 20 months is $\$760.00$

d $\$7550$ over 9 years is $\$6523.20$.

e $\$5800$ over 6 quarters is $\$321.90$.

f $\$5500$ over 24 months is $\$385.00$.

11 Find the principal that earned:

- **a** $\$865$ simple interest, at 3% p.a. over 6 years.
- **b** $\$390$ simple interest, at 2.5% p.a. over 4 years.
- **c** $\$1200$ simple interest, at 1% per month over 2 years.
- **d** $\$3260$ simple interest, at 2.5% per week over 6 months.

12 Sally made loan repayments totalling to $\$4320$ on a loan of $\$4000$ over 4 years.

- **a** Calculate the total simple interest charged on the loan.
- **b** Calculate r, the annual simple interest rate.

13 For his investment into government bonds , Buzz was paid simple interest of 9% p.a. Calculate the size of Buzz's initial investment $\$P$, if he earned $\$294.12$ interest after 2 years.

14 If $\$2196$ is invested at 2% p.a. simple interest for 8 years, what simple interest rate r, would earn the same amount of interest in only 5 years?

Express r as a percentage, correct to two decimal places.

15 $\$552$ is invested at a simple interest rate of 5% p.a.

- **a** Calculate the interest that will be earned on the investment in one year.
- **b** Calculate the number of years it will take the investment to grow to $\$828$.

16 $\$1957$ is invested at a simple interest rate of 6% p.a. Calculate the number of years needed for interest of $\$1174.20$ to be earned on the investment.

17 $\$2379.00$ is invested at a simple interest rate of 2% p.a. Calculate the number of quarters needed for interest of $\$23.79$ to be earned on the investment.

18 $\$63\,790$ is invested at a simple interest rate of 8% p.a. Calculate the number of months needed for interest of $\$7654.80$ to be earned on the investment. Round your answer to the nearest month.

19 $\$7050$ is invested at a simple interest rate of 7% p.a. Calculate the number of years needed for it to grow to $\$7950$. Round your answer to two decimal places.